

Practical No 1

Aim: Design and implement algorithms to encrypt and decrypt messages using classical substitution and transposition techniques.

Caesar Cipher

```
def caesar(text, key):
    min = ord('a')
    max = ord('z')
    size = min + max + 1

    result = ""

    for char in text:
        text_char = ord(char)
        cipher = min + ((text_char - min + key) % size)
        result += chr(cipher)

    return result

def decrypt_caesar(text, key):
    min = ord('a')
    max = ord('z')
    size = min + max + 1

    result = ""

    for char in text:
        text_char = ord(char)
        cipher = min + ((text_char - min - key) % size)
        result += chr(cipher)

    return result

plain_text = input("Enter Plain Text : ").lower().strip()
cipher_text = caesar(plain_text, 5)

print("Plain Text : ", plain_text)
print("Cipher Text : ", cipher_text)
print("Decrypt Text : ", decrypt_caesar(cipher_text, 5))
```

```
ashif@ashif:~/Code/cs-sem-5/cyber-security/practical-1$ python3 caesar-cipher.py
Enter Plain Text : secret
Plain Text :  secret
Cipher Text :  xjhwjy
Decrypt Text :  secret
```

Rail Cipher

for key = 2

```
def encrypt(text):
```

```
    even = ""
```

```
    odd = ""
```

```
    # Read the character
```

```
    for i in range(len(text)):
```

```
        if i % 2 == 0:
```

```
            even += text[i]
```

```
        else:
```

```
            odd += text[i]
```

```
    return even + odd
```

```
def decrypt(cipher):
```

```
    text = ""
```

```
    mid = (len(cipher) + 1) // 2
```

```
    even = cipher[:mid]
```

```
    odd = cipher[mid:]
```

```
    for i in range(len(odd)):
```

```
        text += even[i]
```

```
        text += odd[i]
```

```
    if len(even) > len(odd):
```

```
        text += even[-1]
```

```
    return text
```

```
msg = input("Enter message: ")
```

```
enc = encrypt(msg)
```

```
print("Encrypted: ", enc)
```

```
dec = decrypt(enc)
```

```
print("Decrypted: ", dec)
```

```
ashif@ashif:~/Code/cs-sem-5/cyber-security/practical-1$ python3 rail-cipher-maam.py
Enter message: just testing things
Encrypted:  js etn hnsuttsigtig
Decrypted:  just testing things
```

For Key = n

```
def rail_fence_encrypt(text, key):
    if key <= 1:
        return text

    rails = [""] * key

    rail = 0
    direction = 1 # 1 = down, -1 = up

    for char in text:
        rails[rail] += char

        if rail == 0:
            direction = 1
        elif rail == key - 1:
            direction = -1
        rail += direction

    return ".join(rails)

def rail_fence_decrypt(cipher, key):
    if key <= 1:
        return cipher

    # Mark zigzag positions
    pattern = [" " for _ in cipher] for _ in range(key)]

    rail = 0
    direction = 1

    for col in range(len(cipher)):
        pattern[rail][col] = '*'

        if rail == 0:
            direction = 1
        elif rail == key - 1:
            direction = -1
        rail += direction

    # Fill marked positions
    index = 0
    for r in range(key):
        for c in range(len(cipher)):
            if pattern[r][c] == '*':
                pattern[r][c] = cipher[index]
                index += 1

    # Read zigzag
    result = []

    rail = 0
    direction = 1

    for col in range(len(cipher)):
        result.append(pattern[rail][col])

        if rail == 0:
```

```
        direction = 1
    elif rail == key - 1:
        direction = -1

    rail += direction

    return ".join(result)
```

```
plain_text = input("Enter Plain Text : ")
key = int(input("Enter Key : "))
cipher_text = rail_fence_encrypt(plain_text, key)
```

```
print("Plain Text : ", plain_text)
print("Cipher Text : ", cipher_text)
print("Decrypt Text : ", rail_fence_decrypt(cipher_text, 5))
```

```
ashif@ashif:~/Code/cs-sem-5/cyber-security/practical-1$ python3 rails-cipher.py
Enter Plain Text : just testing things with key = 5
Enter Key : 5
Plain Text : just testing things with key = 5
Cipher Text : jtn usiighk5senhste ttgt iy= w
Decrypt Text : just testing things with key = 5
ashif@ashif:~/Code/cs-sem-5/cyber-security/practical-1$ python3 rails-cipher.py
Enter Plain Text : just testing things with key = 7
Enter Key : 7
Plain Text : just testing things with key = 7
Cipher Text : j ugthksnhtetiiiy tnw tsg =7es
Decrypt Text : jgen7wtt hi etik sissgynuh =tt
```